the next few pages sound a bit abstruse. The theory shows, for the first time, that in the Newtonian vision of the world there isn't just something missing: rather, it must be radically modified – in a way that goes completely against common sense. It is the first real leap into the revision of our most intuitive understanding of the world.

The extended present

The theories of Newton and of Maxwell appear to contradict each other in a subtle way. Maxwell's equations determine a velocity: the velocity of light. But Newton's mechanics is not compatible with the existence of a fundamental velocity, because what enters Newton's equations is acceleration, not velocity. In Newton's physics, velocity can only be velocity of something *with respect to something else*. Galileo had underlined the fact that the Earth moves with respect to the Sun, even if we do not perceive this movement, because what we usually term 'velocity' is velocity 'with respect to Earth'. Velocity, we say, is a *relative* concept, that is, there is no meaning to the velocity of an object by itself: the only velocity which exists is the velocity of an object with respect to another object. This is what physics students learned in the nineteenth century, and what they learn today. But if this is so, then the speed of light determined by Maxwell's equations is velocity with respect to what?

One possibility is that there is a kind of universal substratum in relation to which light moves and has its speed. But the predictions of Maxwell's theory seem to be independent of this substratum. The experimental attempts to measure the speed of the Earth with respect to this hypothetical substratum tried at the end of the twentieth century all failed.

Einstein has claimed that he was not put on the right track by any experiments but only by reflecting on the apparent contradiction between Maxwell's equations and Newton's mechanics. He asked himself whether there was a way of rendering Newton's and Galileo's core discoveries and Maxwell's theory consistent.

In doing so, Einstein arrives at a stupefying discovery. To understand it, think of all the past, present and future events (with respect to the moment in which you are reading) and imagine them distributed as in figure 3.1.

Well, Einstein's discovery is that this diagram is incorrect. In reality, things are actually as they are depicted in figure 3.2.

Between the past and the future of an event (for example, between the past and the future for you, where you are, and in the precise moment in which you are reading) there exists an 'intermediate zone', an 'expanded present'; a zone that is neither past nor future. This is the discovery made with special relativity.